

Yi Xia (Sean)

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Applied AI & ML Software Engineer

Open to: Applied AI Engineer · ML Engineer · Computer Vision Engineer · Data Scientist / Applied Scientist · Software Engineer (AI/ML)

Full Work Rights · No Sponsorship Required

Auckland-based applied AI engineer and PhD candidate who builds **production computer-vision systems**. Four years at **TOHU MEDIA** designing and deploying object detection, OCR and action-recognition pipelines on real film-industry video, with PyTorch training, Docker-packaged inference on cloud GPU servers, and direct iteration with non-technical users. Concurrently completing an AUT PhD in deep learning for human activity recognition; **two first-author Q1 journal papers** (Pattern Recognition and Computer Science Review), seven peer-reviewed papers and 70+ citations. Strong fit for New Zealand teams needing practical AI, ML engineering, video analytics, document/OCR AI, agritech, healthcare, insurance, transport or enterprise data-science work.

PRODUCTION AI 3 shipped CV/OCR/action models in live media workflows.	ENGINEERING Python, PyTorch, Docker, Linux, cloud GPU inference.	RESEARCH DEPTH 2 Q1 first-author papers, PhD, 7 papers, 70+ citations.	NZ READY Auckland-based, local industry experience, full work rights.
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CORE EXPERIENCE

Artificial Intelligence Developer Feb 2022 - present
TOHU MEDIA, Auckland - full-time since June 2025; part-time alongside master's and PhD study before that

- Own the company's AI capability end-to-end: scoped datasets, trained/evaluated models in PyTorch, packaged inference in Docker, and deployed to company cloud GPU servers.
- Designed and shipped **three production AI systems** - object detection, OCR and action recognition - replacing previously manual transcription work on client video footage (specific deliverables under NDA).
- Built for messy real footage: variable cinematography, dense overlay text, low-contrast archival material, long-form takes, and workloads that exceed single-workstation GPU limits.
- Made independent calls on architecture, labelling strategy, post-processing, latency/accuracy trade-offs, and scope reduction under deadline.
- Worked directly with editors and the creative director to define useful model output inside a real production workflow, preserving client data sovereignty on company-controlled infrastructure.

PhD Researcher - Computer Vision & Human Activity Recognition Oct 2023 - Oct 2026 expected
Institute of Robotics & Vision, Auckland University of Technology

- Designed FATL-GCN, a frequency-aware spatio-temporal graph network for skeleton-based human activity recognition; published in **Pattern Recognition** (Q1, IF 7.6) in 2026.
- First-authored a survey on generalisation in skeleton-based HAR, published in **Computer Science Review** (Q1 review journal, IF = 17.5) in 2026, plus Dynamic Self-Attention Gated ST-GCN at ICONIP 2024 - full ownership from idea and model design through GPU training, ablations, writing and rebuttal.
- Mentor master's students on PyTorch, reproducible experiments, research writing and model evaluation.

TECHNICAL CORE

- Programming & ML**
Python, Java, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, pandas, Hugging Face
- Computer Vision**
YOLO family, object detection/tracking, OCR, video analytics, action recognition, segmentation, classification
- Deployment**
Docker, containerised inference, Linux, Git/GitHub, CUDA, cloud GPU servers, model evaluation
- Research methods**
GCNs, CNNs, attention/Transformers, skeleton-based HAR, frequency-domain learning
- Generative AI**
Claude/GPT API use, LLM integration, agentic workflows, function/tool calling, prompt engineering

DOMAIN FIT

- Enterprise AI
- Computer Vision
- OCR / Document AI
- Video Analytics
- Agritech
- Healthcare AI
- Insurance Claims AI
- Transport / Safety
- Applied Research

ADDITIONAL EXPERIENCE

Research Assistant - Computer Vision Nov 2022 - Jul 2023
School of Engineering, Computer & Mathematical Sciences, AUT

- Co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video; lead paper has 50+ citations.
- Co-authored three peer-reviewed papers across IVCNZ 2022, SAI Intelligent Systems 2023 and PSIVT 2023.

Teaching Assistant 2022 - present
Auckland University of Technology

- TA on undergraduate computer-science courses including programming, AI and computer vision; run labs, mark assessments and mentor final-year projects.

Tech Service Apr - Dec 2023
PB Tech, Auckland

- Handled hardware diagnostics, software troubleshooting, customer returns and custom PC builds; strengthened plain-English technical communication with non-specialist users.

Junior Lecturer / Co-Founder 2015 - 2021
Baiyin Vocational & Technology College / Baiyin Pinuo Training School, China

- Taught Computer Fundamentals, Java, IoT applications and RFID technologies to about 120 students per year for four academic years.
- Co-founded a private training school; built curriculum, ran academic operations and finance, then divested before relocating to New Zealand.

SELECTED PUBLICATIONS — 7 PEER-REVIEWED · 70+ CITATIONS

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Beyond Benchmark Accuracy: A Generalization-Centered Analysis of Deep Learning for Skeleton-Based Human Activity Recognition." *Computer Science Review*. Q1 review journal, IF = 17.5.
doi.org/10.1016/j.cosrev.2026.101030

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition." *Pattern Recognition*. Q1 journal, IF 7.6.
doi.org/10.1016/j.patcog.2026.113146

Y. Xia, M. Nguyen, W. Q. Yan (2022). "A Real-Time Kiwifruit Detection Based on Improved YOLOv7." *IVCNZ 2022*, Springer LNCS. 50 citations.
doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). "Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR." *ICONIP 2024*, Springer LNCS.
doi.org/10.1007/978-981-96-6588-4_14 · full list on Google Scholar

Practical AI engineer with research depth: production computer vision, end-to-end ML deployment, and New Zealand work readiness.

EDUCATION

PhD, Computer & Information Sciences
Auckland University of Technology
Oct 2023 - Oct 2026 expected. AUT PhD Full Scholarship. Thesis: deep learning for skeleton-based human activity recognition.

Master of Computer & Information Sciences
Auckland University of Technology
2021 - 2023. Specialisation: Computer Vision.

BEng, Logistics Engineering (Software)
Dalian Neusoft University of Information
2011 - 2015.

WHY THIS FITS NZ EMPLOYERS

- Skilled tech role fit:** AI/ML engineering, software engineering for AI systems, computer vision, data-science engineering and applied research.
- Local context:** Auckland-based since 2021 with NZ industry experience across TOHU MEDIA and PB Tech.
- Evidence of delivery:** production models, Docker deployment, cloud GPU inference, user-facing workflow integration.
- Evidence of depth:** AUT PhD scholarship, two Q1 first-author journal papers, seven peer-reviewed publications, 70+ citations.

AWARDS & PROFESSIONAL SERVICE

- AUT PhD Full Scholarship** - tuition + living stipend, Oct 2023 - Oct 2026.
- SECMS Research Assistant Fund** and **AUT Summer Research Assistant Scholarship**.
- Peer reviewer for ACM TOMM, International Agrophysics and IEEE PIMRC.
- Member, AUT Institute of Robotics & Vision.

COMMUNICATION

- Eight years of cumulative teaching experience across China and New Zealand.
- Comfortable explaining AI systems to engineers, students, editors, managers and non-specialist stakeholders.
- Bilingual: Mandarin Chinese and professional English.

REFEREES

Available on request. Doctoral supervisors at AUT and an industry referee from TOHU MEDIA can be provided.